ARTIN-REES CONTROLS THE A-ADIC STABILISATION OF THE HOMOLOGY OF A COMPLEX OF FINITE MODULES OVER A NOETHERIAN RING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author writes 'We omit the verification that the diagram commutes.' in the counted proof span; that sentence is kept verbatim and is the one step left to the reader. The author's typo 'the cass of z' in the last paragraph is reproduced unchanged.

Editorial note. The three cross-chapter references inside the counted proof span (algebra-lemma-map-AR, algebra-lemma-Artin-Rees and categories-remark-pro-category) are recorded as non-load-bearing navigation pointers; both Artin-Rees applications and everything resting on them stay inside the counted proof span.

Allowed standard prerequisites (editorial).

- The Artin-Rees lemma for an ideal in a Noetherian ring. applied at proof lines 28916-28920 through the cited Algebra lemmas. Supplies the two constants c_1 and c_2 whose sum c controls every subsequent statement.
- Pro-objects of modules and their morphisms. the isomorphism claim in part (1) at statement lines 28894-28895 and the cited Categories remark at proof lines 28962-28963. Fixes what it means for (H_n) and (H/I^n H) to be isomorphic, which is the primary conclusion.
- Inverse limits of modules and Mittag-Leffler conditions. parts (2) and (3) at statement lines 28895-28896. Makes sense of the limit comparison and of the stabilization of images claimed in part (4).

Source reference. Original lemma 00IN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 685–685. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00IO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 686–686. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

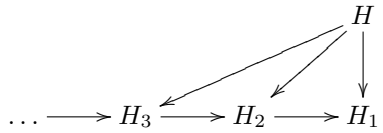
Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 05PX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7356–7356. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let I be an ideal of a Noetherian ring A. Let $K \xrightarrow{\alpha} L \xrightarrow{\beta} M$ be a complex of finite A-modules. Set $H = \mathrm{Ker}(\beta)/\mathrm{Im}(\alpha)$. For $n \geq 0$ let*

$$K/I^nK \xrightarrow{\alpha_n} L/I^nL \xrightarrow{\beta_n} M/I^nM$$

be the induced complex. Set $H_n = \mathrm{Ker}(\beta_n)/\mathrm{Im}(\alpha_n)$. Then there are canonical A-module maps giving a commutative diagram



Moreover, there exists a $c > 0$ and canonical A-module maps $H_n \to H/I^{n-c}H$ for $n \geq c$ such that the compositions

$$H/I^nH \to H_n \to H/I^{n-c}H \quad \text{and} \quad H_n \to H/I^{n-c}H \to H_{n-c}$$

are the canonical ones. Moreover, we have

1. *(H_n) and (H/I^nH) are isomorphic as pro-objects of Mod_A,*
2. *$\lim H_n = \lim H/I^nH$,*
3. *the inverse system (H_n) is Mittag-Leffler,*
4. *the image of $H_{n+c} \to H_n$ is equal to the image of $H \to H_n$,*
5. *the composition $I^cH_n \to H_n \to H/I^{n-c}H \to H_n/I^{n-c}H_n$ is the inclusion $I^cH_n \to H_n$ followed by the quotient map $H_n \to H_n/I^{n-c}H_n$, and*
6. *the kernel and cokernel of $H/I^nH \to H_n$ is annihilated by I^c.*

Proof. Observe that $H_n = \beta^{-1}(I^nM)/\mathrm{Im}(\alpha)+I^nL$. For $n \geq 2$ we have $\beta^{-1}(I^nM) \subset \beta^{-1}(I^{n-1}M)$ and $\mathrm{Im}(\alpha)+I^nL \subset \mathrm{Im}(\alpha)+I^{n-1}L$. Thus we obtain our canonical map $H_n \to H_{n-1}$. Similarly, we have $\mathrm{Ker}(\beta) \subset \beta^{-1}(I^nM)$ and $\mathrm{Im}(\alpha) \subset \mathrm{Im}(\alpha) + I^nL$ which produces the canonical map $H \to H_n$. We omit the verification that the diagram commutes.

By Artin-Rees we may choose $c_1, c_2 \geq 0$ such that $\beta^{-1}(I^nM) \subset \mathrm{Ker}(\beta) + I^{n-c_1}L$ for $n \geq c_1$ and $\mathrm{Ker}(\beta) \cap I^nL \subset I^{n-c_2}\,\mathrm{Ker}(\beta)$ for $n \geq c_2$, see Algebra, Lemmas 00IO and 00IN. Set $c = c_1 + c_2$.

Let $n \geq c$. We define $\psi_n : H_n \to H/I^{n-c}H$ as follows. Say $x \in H_n$. Choose $y \in \beta^{-1}(I^nM)$ representing x. Write $y = z + w$ with $z \in \mathrm{Ker}(\beta)$ and $w \in I^{n-c_1}L$ (this is possible by our choice of c_1). We set $\psi_n(x)$ equal to the class of z in $H/I^{n-c}H$. To see this is well defined, suppose we have a second set of choices y', z', w' as above for x with obvious notation. Then $y' - y \in \mathrm{Im}(\alpha) + I^nL$, say $y' - y = \alpha(v) + u$ with $v \in K$ and $u \in I^nL$. Thus

$$y' = z' + w' = \alpha(v) + u + z + w \Rightarrow z' = z + \alpha(v) + u + w - w'$$

Since $\beta(z' - z - \alpha(v)) = 0$ we find that $u + w - w' \in \mathrm{Ker}(\beta) \cap I^{n-c_1}L$ which is contained in $I^{n-c_1-c_2}\,\mathrm{Ker}(\beta) = I^{n-c}\,\mathrm{Ker}(\beta)$ by our choice of c_2. Thus z' and z have the same image in $H/I^{n-c}H$ as desired.

The composition $H/I^nH \to H_n \to H/I^{n-c}H$ is the canonical map because if $z \in \mathrm{Ker}(\beta)$ represents an element x in $H/I^nH = \mathrm{Ker}(\beta)/\mathrm{Im}(\alpha) + I^n\,\mathrm{Ker}(\beta)$ then it is clear from the above that x maps to the class of z in $H/I^{n-c}H$ under the maps constructed above.

Let us consider the composition $H_n \to H/I^{n-c}H \to H_{n-c}$. Given x, y, z, w as in the construction of ψ_n above, we see that x is mapped to the cass of z in H_{n-c}. On the other hand, the canonical map $H_n \to H_{n-c}$ from the first paragraph of the proof sends x to the class of y. Thus we have to show that $y - z \in \mathrm{Im}(\alpha) + I^{n-c}L$ which is the case because $y - z = w \in I^{n-c_1}L \subset I^{n-c}L$.

Statements (1) – (4) are formal consequences of what we just proved. Namely, (1) follows from the existence of the maps and the definition of morphisms of pro-objects in Categories, Remark 05PX. Part (2) holds because isomorphic pro-objects have isomorphic limits. Part (3) is immediate from part (4). Part (4) follows from the factorization $H_{n+c} \to H/I^nH \to H_n$ of the canonical map $H_{n+c} \to H_n$.

Proof of part (5). Let $x \in I^cH_n$. Write $x = \sum f_i x_i$ with $x_i \in H_n$ and $f_i \in I^c$. Choose y_i, z_i, w_i as in the construction of ψ_n for x_i. Then for the computation of ψ_n of x we may choose $y = \sum f_i y_i$, $z = \sum f_i z_i$ and $w = \sum f_i w_i$ and we see that $\psi_n(x)$ is given by the class of z. The image of this in $H_n/I^{n-c}H_n$ is equal to the class of y as $w = \sum f_i w_i$ is in I^nL. This proves (5).

Proof of part (6). Let $y \in \mathrm{Ker}(\beta)$ whose class is x in H. If x maps to zero in H_n, then $y \in I^nL + \mathrm{Im}(\alpha)$. Hence $y - \alpha(v) \in \mathrm{Ker}(\beta) \cap I^nL$ for some $v \in K$. Then $y - \alpha(v) \in I^{n-c_2}\,\mathrm{Ker}(\beta)$ and hence the class of y in H/I^nH is annihilated by I^{c_2}. Finally, let $x \in H_n$ be the class of $y \in \beta^{-1}(I^nM)$. Then we write $y = z + w$ with $z \in \mathrm{Ker}(\beta)$ and $w \in I^{n-c_1}L$ as above. Clearly, if $f \in I^{c_1}$ then fx is the class of $fy + fw \equiv fy$ modulo $\mathrm{Im}(\alpha) + I^nL$ and hence fx is the image of the class of fy in H as desired. □